Behavioral Dynamics and Alternative Data for Thin-File Credit Default Prediction

DATASCI 207: Applied Machine Learning at Scale | Project Milestone Report

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GitHub Repository: https://github.com/romanmhall/DATASCI-207-FinalProject-Team1

1. Motivation & Research Question

Traditional lending relies on credit bureau histories (FICO, VantageScore). Applicants without multi-year credit lines are classified as “thin-file” or “unscoreable” and are often rejected or charged high rates, even when they earn steadily and manage cash well; this mostly affects young adults, recent immigrants, and informal economy workers. Berg et al. (2020) found that digital footprints predict default with an AUC of 0.696, rising to 0.736 when combined with bureau data.

We ask whether behavioral history (installment payment delays, previous application outcomes, and POS cash balances) can replace or add to missing bureau records when predicting default. We test two hypotheses: (H1) a hybrid feature set outperforms traditional features alone, and (H2) the gain is largest for applicants with no bureau history. The target is binary: y = 1 if the client paid late by more than a set number of days on at least one of the loan’s first installments (Home Credit’s definition of payment difficulties), and y = 0 otherwise.

2. Dataset Architecture & Preprocessing Pipeline

We use the Home Credit Default Risk dataset from Kaggle: a primary application table joined to four relational tables of past credit behavior.

Dataset / Table	Raw Records	Raw Size	Processed Size	Key Extracted Features / Aggregations
application_train.csv	307,511	325.2 MB	84.1 MB (-74.1%)	Demographics, income, credit, annuity, external scores
bureau.csv	1,716,428	92.7 MB	42.9 MB (-53.7%)	Active and total lines, overdue debt, max DPD
previous_application.csv	1,670,214	66.6 MB	35.1 MB (-47.7%)	Refusal count, approval rate, down payment ratio
POS_CASH_balance.csv	10,001,358	462.4 MB	100.0 MB (-78.4%)	Contract status counts, balance trajectory, max DPD
installments_payments.csv	13,605,401	544.2 MB	381.0 MB (-30.0%)	Payment delay (days past due), underpayment frequency
Consolidated Matrix (processed_features.parquet)	**307,511**	**1,491.1 MB (Raw)**	**37.3 MB (Snappy)**	**153 features, downcast, left-joined on SK_ID_CURR**

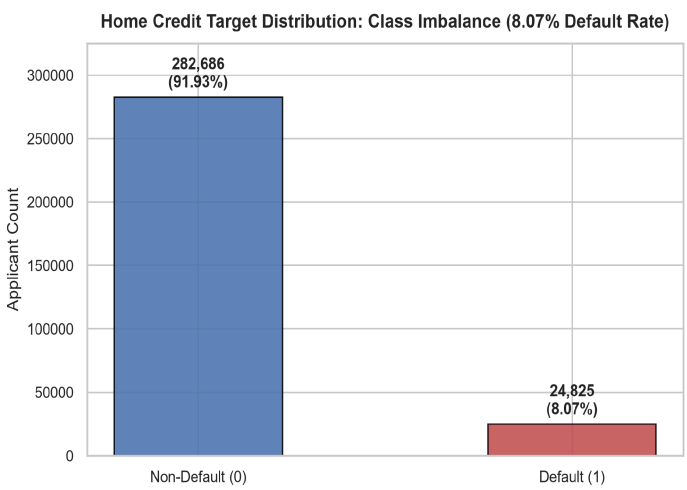
Preprocessing & Multi-Table Relational Pipeline

1. ***Target Imbalance:*** The target has 282,686 non-defaults (91.93%) and 24,825 defaults (8.07%), an 11.39:1 ratio.
2. ***Missing Values:*** 67 of the 122 raw application features have gaps and 41 are more than 50% missing; we keep them rather than drop them. EXT_SOURCE_1 is 56.4% missing, EXT_SOURCE_2 0.2%, and EXT_SOURCE_3 19.8%. Numeric gaps take the training-set median and categorical gaps a separate “Missing” category.
3. ***Outlier Treatment:*** 55,374 applicants (18.01%) have DAYS_EMPLOYED = 365,243, a placeholder used for pensioners and people not in work. We flag them (DAYS_EMPLOYED_ANOM = 1), recode the value to NaN, and impute it with the training median.
4. ***Feature Engineering:*** PAYMENT_RATE (AMT_ANNUITY / AMT_CREDIT), INCOME_CREDIT_RATIO (AMT_INCOME_TOTAL / AMT_CREDIT), and ANNUITY_INCOME_RATIO (AMT_ANNUITY / AMT_INCOME_TOTAL).
5. ***Relational Aggregation:*** The four behavioral tables (about 27.0M rows) are aggregated per SK_ID_CURR and left-joined to the application table. Missing counts become 0 (no history); rates such as mean payment delay stay NaN so they do not imply perfect repayment. HAS_BUREAU_HISTORY = 1 marks applicants with at least one bureau credit line.
6. ***Transformation, Encoding & Scaling:*** 70 right-skewed, non-negative features are log-transformed (log1p where zeros occur), categoricals are one-hot encoded (up to 289 inputs), and all inputs are standardized with a

StandardScaler fit on the training split. Feature selection is done by cohort (Section 5) rather than by an automated method.

7. ***Dataset Splitting:*** A stratified 80/20 split gives 246,008 training rows (19,860 defaults) and 61,503 held-out test rows (4,965 defaults), both 8.07% default. 5-fold stratified cross-validation within the training set (about 196,800 / 49,200 rows per fold) serves as validation and sets each model's decision threshold.

3. Exploratory Data Analysis & Visual Insights



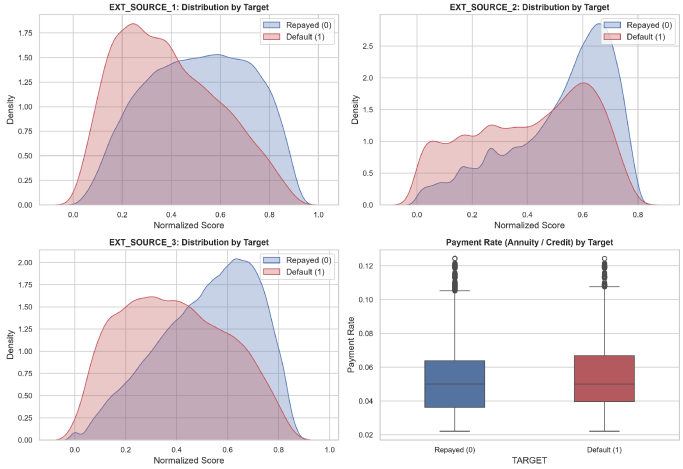


Figure 1: Left: Extreme target class imbalance (8.07% default rate, 11.39:1 non-default to default ratio). Right: Top 25 feature missingness proportions, highlighting sparse external risk signals.

Figure 2: Density curves (KDE) for external bureau scores (EXT_SOURCE_1/2/3) by target status, alongside payment rate boxplots. Defaulters exhibit systematically lower bureau scores and higher payment annuity-to-credit ratios.

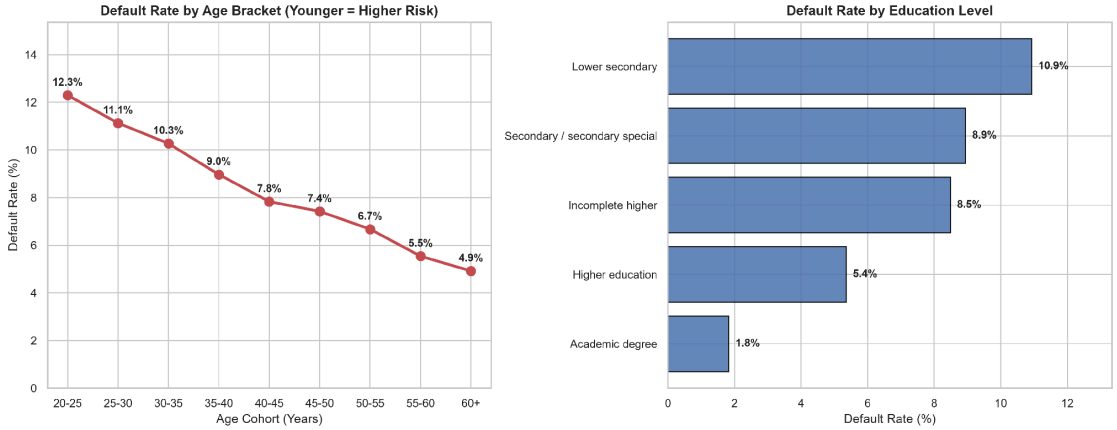


Figure 3: Demographic risk gradients. Default rates drop monotonically from 12.3% for borrowers aged 20-25 down to 5.2% for borrowers aged 60+, validating that younger, thin-file cohorts bear heightened default risk.

Key EDA Insights

- EXT_SOURCE_3 (r = −0.1789), EXT_SOURCE_2 (r = −0.1605), and EXT_SOURCE_1 (r = −0.1553) are the features most correlated with default.
- Applicants with the DAYS_EMPLOYED placeholder default at 5.40% versus 8.66% for others, so the flag carries signal.
- SHAP values for the hybrid LightGBM model assign 58% of importance to traditional features, 36% to behavioral features and external scores, and 6% to the ratios.

4. Engineering Challenges & Scaling Solutions

1. **Relational Table Scale:** The 27.0M relational rows did not fit in Colab's 12.7 GB of RAM. We loaded only needed columns, downcast numeric types and cast low-cardinality strings to category (30% to 78.4% less memory per

table), processed one table at a time with explicit garbage collection, and saved the joined matrix once as Snappy Parquet (37.3 MB).

2. **Temporal Leakage Prevention:** DAYS_* and MONTHS_BALANCE are offsets from the application date. Their maxima are 0 (bureau DAYS_CREDIT) and −1 (installment and POS_CASH dates), so every behavioral record predates the application.
3. **Class Imbalance (11.39:1):** We weight the positive class (scale_pos_weight = 11.39), tune each decision threshold for F1 on out-of-fold predictions, and compare models on ROC-AUC and PR-AUC rather than accuracy.

5. Methods & Planned Experiments

Model Tier	Model Architecture	Key Hyperparameters & Tuning Strategy	Role & Comparison Method
Baseline	Regularized Logistic Regression & Shallow Decision Tree	L2 (C = 0.1) and ElasticNet (L1 ratio 0.5) logistic regression; decision tree (depth 8, 100+ rows per leaf)	Interpretable benchmark that each improvement must beat.
Improvement 1	LightGBM Gradient Boosted Decision Trees	scale_pos_weight = 11.39; now 400 trees, learning rate 0.1, 31 leaves; planned grid search over learning rate (0.01 to 0.05) and leaves (31 to 63)	Captures non-linear interactions; SHAP values give feature importance by source.
Improvement 2	Deep Tabular Neural Network (TensorFlow / Keras)	Entity embeddings, dense residual layers, dropout 0.25, batch norm, Adam (learning rate 0.001), early stopping on validation ROC-AUC	Tests whether learned embeddings improve on tree ensembles for tabular data.

Evaluation Metrics & Validation Strategy

- **Validation:** 5-fold stratified CV on the 246,008 training rows for tuning and thresholds; each final model is refit on all training rows and scored once on the same 61,503 held-out rows.
- **Metrics:** ROC-AUC (ranking), PR-AUC (minority class), F1 at the tuned threshold, and Brier score after calibration, since class weighting inflates probabilities. The best model per cohort has the highest held-out ROC-AUC, checked against PR-AUC.
- **Feature Cohorts:** 1 (traditional) is static application fields plus bureau aggregates; 2 (alternative) is behavioral aggregates plus EXT_SOURCE scores; 3 (hybrid) is both plus the ratios; 4 is static, behavioral, and ratio features without bureau data.
- **Subgroup Hypothesis Testing (H2):** We split the held-out rows by HAS_BUREAU_HISTORY into Group A (bureau records) and Group B (none, about 14% of applicants) and bootstrap the hybrid-over-traditional AUC gain for each group.
- **Preliminary Results:** Held-out ROC-AUC on Cohorts 1 to 4 is 0.705, 0.730, 0.760, and 0.713 for L2 logistic regression and 0.713, 0.731, 0.770, and 0.727 for LightGBM. The hybrid cohort beats the traditional one by about 0.055 AUC for both models, supporting H1; hybrid LightGBM is best so far (PR-AUC 0.261). Cohort 2 includes the EXT_SOURCE external credit scores, so we will rerun it without them to isolate the behavioral features.

6. Team Contributions & Notebook Ownership Log

Each member owns one notebook in the repository, with cell numbering starting at 1.

Team Member	Email Address	Owned Notebook / Track	Core Contributions
Roman Hall	romanhall@berkeley.edu	01_eda_home_credit.ipynb	Data cleaning, missingness profiling, DAYS_EMPLOYED repair, ratio features, EDA figures
Tung La	tung_la@berkeley.edu	02_eda_relational_tables.ipynb	Relational aggregation, memory optimization, Parquet export; Sections 2 and 4
Faisal Khasawneh	faisal_k@berkeley.edu	03_alternative_features.ipynb	Feature cohorts, baselines, LightGBM with 5-fold CV, SHAP analysis
Advik Goel	advik_goel@berkeley.edu	04_model_experiments.ipynb	Report lead and Section 5; planned TensorFlow model and subgroup analysis